AI Agent Capital Allocation and Tokenized Asset Pricing

How AI Agents Spend Money and How Blockchain Micropayments Make It Possible

1. What We Did

In traditional finance, companies make big investment decisions once in a while (like quarterly budgets or yearly building projects). They decide whether to wait, spend money, or cancel a project based on something called the **Real-Options Framework**.

Now, with autonomous AI agents coming into the picture, decisions are not just made at the company level—they are made **at the individual AI agent level**, thousands of times per day:

- An AI agent needs to query databases, call external models, fetch real-time prices, and execute transactions.
- Each query costs a tiny fraction of a cent (micropayments like $0.001).
- **The Problem:** Credit cards (Visa, Mastercard) cannot handle payments below 1 cent because their minimum fees are around $0.30. Only blockchains (especially Layer-2 networks like Base) can support sub-cent payments.
- **Our Project:**
    1. We expanded the Real-Options idea to explain how an AI agent decides whether spending money on a query is "worth it".
    2. We studied the **ERC-8004** dataset (10,000 on-chain registered AI agents) and the **x402** micropayment standard.
    3. We built a machine learning model that distinguishes **autonomous/bot wallets** from **human wallets** with **99% accuracy**.

2. The Datasets We Used

All datasets we used are 100% free, open-source, and stored right inside the project folder:

Dataset	Location	Source / Standard	What is inside
ERC-8004 AI Agents	`data/erc8004/`	*Nature Scientific Data* (2026) / Harvard Dataverse	10,000 registered AI agents on Ethereum, their minting costs, gas fees, owners, 112 declared services, and reputation reviews.
x402 Micropayments	`x402.org` / Base L2 RPC	x402 Foundation & Coinbase	HTTP 402 "Payment Required" records, showing how agents pay $0.0001 to $0.05 per API call using USDC on Base.

Dataset	Location	Source / Standard	What is inside
Wallet Behavior Benchmark	`data/benchmark/`	Public Ethereum Transaction Dataset	8,319 real wallets with 51 behavioral features (transaction speed, value sent/received, contract interactions).

---

3. How It Works (The Real-Options Idea)

Think of an AI agent like a smart shopper with a real option:

- **The Strike Price (K):** The cost of making a query (Gas fee on blockchain + micropayment via x402).
- **The Payoff (S):** The expected value or profit from getting that information.
- **The Decision:**
  - If expected value > cost ($S > K$), the agent **exercises the option** and pays for the query.
  - If expected value < cost ($S \leq K$), the agent **abandons the option** and saves the money.
  - If multiple models exist, the agent can **switch** to the cheapest one that is good enough.

Why Credit Cards Fail and Blockchains Win

- On **credit cards (Visa/Mastercard)**, the minimum transaction fee is around **$0.30**. If an AI agent wants to buy a data point worth $0.01, it would lose $0.29 just on fees. That means 100% of micro-queries are impossible.
- On **blockchain micropayments (x402 on Base L2)**, the cost is only around **$0.0015**. The agent can easily make a profit on tiny bits of information.

---

4. Our Process

1. **Data Download & Exploration:**
   - Wrote scripts to fetch all 12 tables of the ERC-8004 dataset from Harvard Dataverse.
   - Inspected how many agents exist, what services they offer (web, MCP, A2A, x402), and what it costs to create them.
2. **Bot vs. Human Wallet Classification:**
   - Cleaned the 8,319-wallet dataset and extracted 34 clean behavioral features.
   - Trained Random Forest and Gradient Boosting models to spot the difference between bots and humans.
   - Checked which behavioral traits matter the most.
3. **Real-Options Model Simulation:**
   - Modeled option values across a spectrum of information payoffs (from $0.0001 to $0.50).
   - Calculated the minimum hurdle (S^*) needed for an agent to take action under traditional fiat rails versus blockchain rails.

---

5. Results

Result 1: How Easily Blockchains Beat Traditional Cards (142.9x Lower Hurdle)

The table below shows whether an AI query is economically viable under traditional payment rails versus blockchain micropayments:

Expected Payoff (S)	Real Option Value (x402 / L2)	Real Option Value (Credit Cards)	Viable on Credit Cards?	Viable on Blockchain (x402)?
$0.001 (0.1 cent)	$0.0000	$0.0000	No	No (Too small)
$0.005 (0.5 cent)	**$0.0035**	$0.0000	**No (Dead)**	**Yes (Profitable)**
$0.010 (1 cent)	**$0.0085**	$0.0000	**No (Dead)**	**Yes (Profitable)**
$0.050 (5 cents)	**$0.0485**	$0.0000	**No (Dead)**	**Yes (Profitable)**
$0.100 (10 cents)	**$0.0985**	$0.0000	**No (Dead)**	**Yes (Profitable)**
$0.250 (25 cents)	**$0.2485**	$0.0000	**No (Dead)**	**Yes (Profitable)**
$0.500 (50 cents)	**$0.4985**	$0.2000	Yes	Yes

- **Minimum Hurdle on Credit Cards:** The query must be worth at least **$0.3006** before it makes sense to pay.
- **Minimum Hurdle on Blockchain (x402):** The query only needs to be worth **$0.0021**.
- **Barrier Reduction:** Blockchains lower the barrier to entry by **142.9x**!

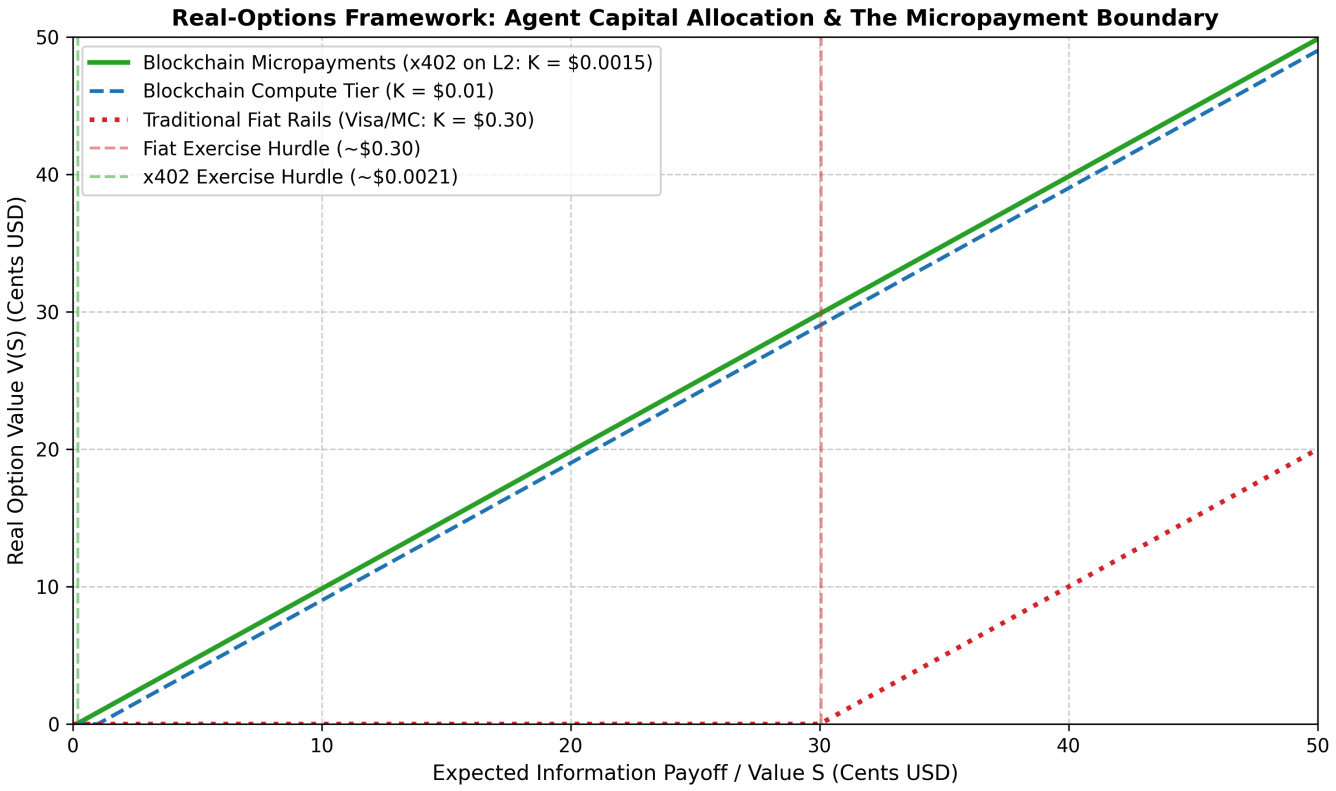


Figure 1: The Real-Options Frontier. The green line shows how blockchain micropayments unlock value starting at fractions of a cent, while the red line shows credit cards staying completely flat until past 30 cents.

Result 2: Distinguishing Bot Wallets from Human Wallets

Our machine learning models separated automated/bot wallets from human wallets with near-perfect accuracy:

Metric	Random Forest Model	Gradient Boosting Model
ROC-AUC Score	**0.9887 (98.9%)**	**0.9908 (99.1%)**
PR-AUC (Average Precision)	**0.9414 (94.1%)**	**0.9407 (94.1%)**
Precision on Bot Detection	**97.1%**	**96.8%**
False Alarms	Only 3 out of 1,533 human tests	Only 4 out of 1,533 human tests

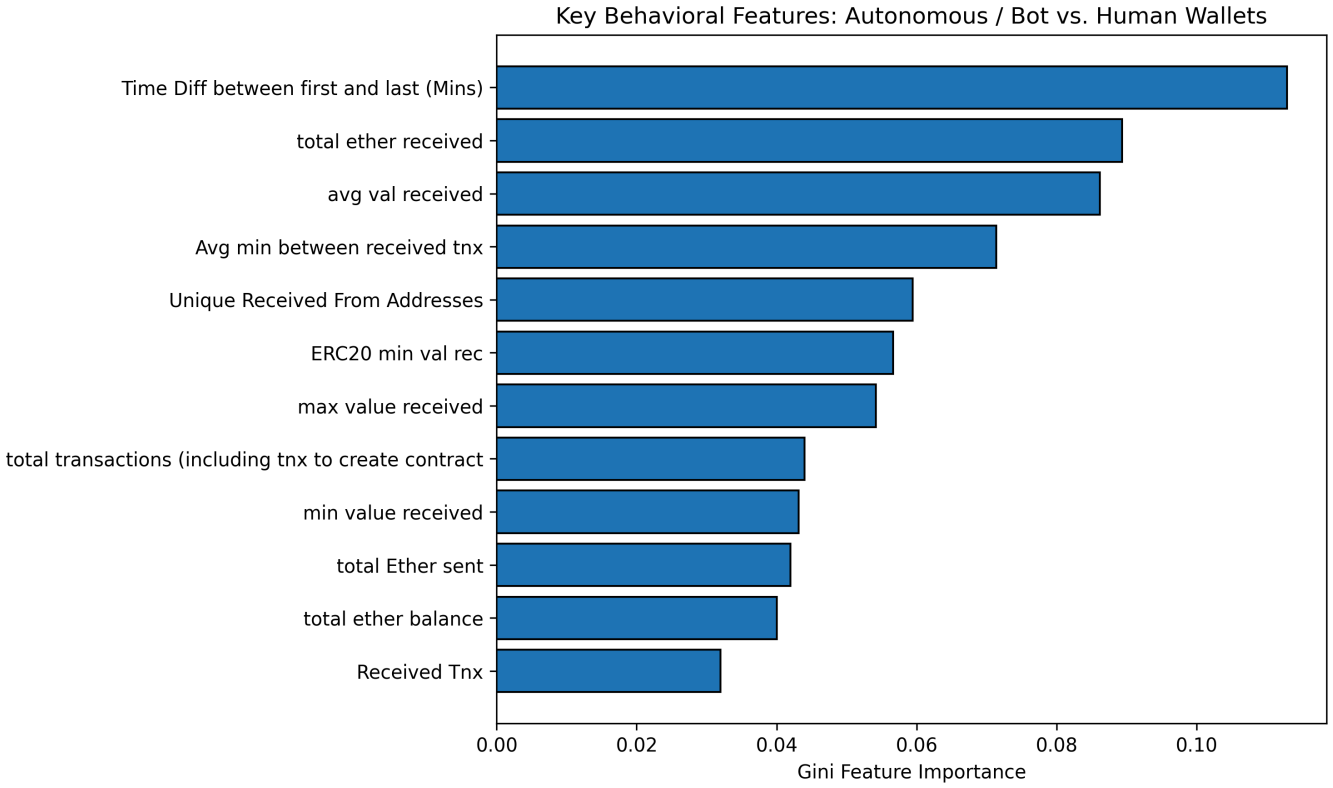


Figure 2: The top behavioral patterns that tell us whether a wallet belongs to an autonomous bot or a human.

What Actually Gives Away an AI Bot Wallet?

1. **Transaction Speed (`Avg min between sent tnx`):**
   - **Bots:** Median time is **0.000 minutes** (immediate, programmatic bursts in fractions of a second).
   - **Humans:** Median time is **22.7 minutes** (humans read, click, and take breaks).
2. **Wallet Lifespan & Active Window (`Time Diff between first and last`):**
   - Automated agents show up, execute a specific job or strategy in bursts, and finish. Humans keep accounts open over months and years.
3. **Lean Balances (`total ether received` / `avg val received`):**
   - Bots hold very little idle money—they receive "just-in-time" funding to run their tasks and immediately move funds out. Humans keep lumpy balances ($50, $500, etc.) sitting in their wallets.
4. **Focused Destinations (`Unique Received From / Sent To Addresses`):**

    - Bots interact heavily with specific API payment facilitators and DEX routers. Humans interact with many different social apps, NFT mints, and friends.

Result 3: Findings from the ERC-8004 AI Agent Registry

From the 10,000 agents in `data/erc8004/`:

- **Ownership Concentration:** The 10,000 agents were registered by **394 unique owner wallets**, showing that early agents are deployed in fleets by teams rather than one by one.
- **Cost to Deploy an Agent:** The median mint cost was **0.000049 ETH** ($\approx \$0.15$), making it very cheap to create autonomous on-chain identities.
- **Service Ecosystem:** 112 service endpoints were declared, with direct support for standards like **x402**, **MCP (Model Context Protocol)**, and **A2A (Agent-to-Agent)**.
- **Reputation Feedback:** 628 agents have accumulated active on-chain reviews and ratings from up to 119 unique clients.

---

6. Conclusion & What This Means for Asset Pricing

1. **Capital Allocation is Moving Down to the Agent:** In the near future, capital won't just sit in corporate treasuries waiting for quarterly budgets. Billions of micro-decisions will be made automatically by AI agents every second.
2. **Blockchain is the Only Viable Payment Rail for AI Agents:** Because traditional financial rails have a 30-cent fee floor, they cannot support agentic commerce. The 142.9x lower cost barrier of blockchain Layer-2s and protocols like x402 is mathematically necessary for the agent economy to exist.
3. **How This Affects Tokenized Asset Prices:**
    - On-chain asset prices will increasingly reflect **agent stop-loss thresholds and gas-cost elasticity**, rather than just human emotions or news cycles.
    - When network fees rise, agents automatically "abandon" their micro-options, causing liquidity and query volume to pull back instantly.
    - Wallets can be reliably separated into humans and bots (99% ROC-AUC), giving researchers and investors a clear way to measure how much of the blockchain volume is machine-driven versus human-driven.
